1) $T(s) = \dfrac{16}{s^2+3s+16}$

$T_s(s) = \dfrac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$

$\omega_n^2 = 16$

$\omega_n = \sqrt{16} = 4$

2) $2\zeta\omega_n = 3$

$\zeta = 3/8$

$\zeta = 0{,}375$

3) Subamortecido

4)

$T_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \dfrac{\pi}{4\sqrt{1-0{,}375^2}} = 0{,}847\ s$

$T_s = \dfrac{4}{\zeta\omega_n} = \dfrac{\cancel{4}}{0{,}375\cdot\cancel{4}} = \dfrac{1}{0{,}375} = 2{,}666 = 2{,}67\ s$

5) dado $\zeta = 0{,}375$,

$\%OP = 100\cdot e^{-(\zeta\pi/\sqrt{1-\zeta^2})}$

$= 28{,}05 = 28\%$

6) $T(s) = \dfrac{100}{s^2+25+100}$

$\omega_n^2 = 100$

$\omega_n = 10$

$T_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

$2\omega_n\zeta = 25$

$\zeta = 1{,}25$

$T_p =$ NÃO TEM, POIS $1-1{,}25^2 < 0$

$T_s = \dfrac{4}{\omega_n\zeta} = \dfrac{4}{10\cdot 1{,}25} = 0{,}32\ s$

7) $\%UP = 12\ \%$

$T_s = 0{,}6\ s$

$$\xi = \frac{-\ln(\%UP/100)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%UP}{100}\right)}}$$

$$\xi = \frac{-\ln(12/100)}{\sqrt{\pi^2 + \ln^2\left(\frac{12}{100}\right)}} = 0{,}5594$$

$$T_s = \frac{4}{\omega_n \xi}$$

$$\omega_n = \frac{4}{T_s \xi} = \frac{4}{0{,}6 \cdot 0{,}5594} = 11{,}94$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} = \frac{\pi}{11{,}94\sqrt{1-0{,}5594^2}} = 0{,}3175$$

$$s^2 + 2\omega_n \xi + \omega_n^2$$

$$s^2 + 13{,}33 s + 142$$

$$s = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$s = -6{,}667 \pm 9{,}878 i$$

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8) $\%UP = 10\%$ $\quad T_p = 5\ s$

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$$\xi = \frac{-\ln(\%UP/100)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%UP}{100}\right)}}$$

$$\xi = \frac{-\ln(10/100)}{\sqrt{\pi^2 + \ln^2\left(\frac{10}{100}\right)}}$$

$$\xi = 0{,}5911$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\omega_n = \frac{\pi}{T_p\sqrt{1-\xi^2}} = 0{,}7789\ s$$

$$s = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$s = -0{,}4604 \pm 0{,}628$$

$$s^2 + 2\omega_n\xi + \omega_n^2$$

$$s^2 + 0{,}9210 s + 0{,}6066$$